
Sequence-Level Accuracy Estimation Under Covariate Shift via Out-of-Distribution Detection Rates for Deep Neural Networks

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Abstract

1 Deep vision models often experience accuracy degradation under covariate shift.
2 This raises challenges with respect to reliability and trustworthiness, as deployment
3 streams typically lack labels to measure this degradation. We study whether
4 aggregate out-of-distribution (OOD) detector verdict rates can serve as sequence-
5 level indicators of network accuracy under shift. Across five benchmarks, multiple
6 architectures, six post-hoc OOD detectors, synthetic perturbations, and naturally
7 occurring domain shifts, we observe that detection rates frequently correlate with
8 generalization gaps. We exploit this relationship by fitting a simple linear model
9 that maps OOD detection rate to accuracy degradation using labeled synthetic
10 calibration shifts. At inference time, the estimator requires only unlabeled inputs.
11 Under a leave-one-shift-out protocol, it achieves mean absolute errors between 0.01
12 and 0.10 and outperforms competing methods for unlabeled accuracy estimation.
13 These results indicate that OOD detectors can provide useful signals for monitoring
14 model performance under covariate shift, with the caveat that their reliability
15 depends on the reliability of the observed alignment between detectability and
16 accuracy degradation.

17 1 Introduction

18 Despite their remarkable performance on benchmarks compared to competing approaches, visual
19 Deep Neural Networks (DNNs) have been shown to frequently fail in real-world scenarios, a phe-
20 nomenon known as *generalization failure* [3, 12, 14, 19, 36, 58]. In essence, these failures arise from a
21 misalignment between the nature of the data on which DNNs is typically trained and evaluated and the
22 broader, more variable data distributions encountered in deployment. This misalignment is commonly
23 described as a *covariate shift* and adversely impacts DNN reliability. Under covariate shifts, DNNs
24 are not guaranteed to yield correct predictions at the rate specified from initial evaluations, which are
25 typically based on In-Distribution (InD) data extracted from the overall training set. These limitations
26 pose serious challenges to the safe deployment of visual DNNs, particularly in high-stakes domains
27 such as medical imaging and autonomous vehicles.

28 Although significant strides have been made with respect to improving the generalization capabilities
29 of DNNs under covariate shift [46], notably through domain-adaptation frameworks [17, 57] and,
30 recently incorporating text corpora [28], attaining sufficient levels of generalization for responsible
31 deployment in high-risk settings remains an open problem. As a result, emerging work is instead
32 focusing on the implementation of support components that can mitigate the risks associated with poor
33 generalization. Notable examples thereof include the implementation of guardrails for Large Lan-
34 guage Models (LLMs) [4, 62] and abstention classes [27, 49]. *Out-of-Distribution (OOD) detectors*
35 have recently also been explored as a means of ensuring output reliability through the operational-

ization of abstention in DNNs [52, 53]. In other work, OOD detectors have been demonstrated to facilitate unlabeled deployment-time accuracy estimation of loss [54] and accuracy [18, 51], albeit requiring labeled distributional shifts on which to calibrate.

In this paper, we study how aggregate OOD detector verdict rates can be used to characterize covariate-shift severity and estimate sequence-level generalization. Our approach is motivated by the works of Vaze et al. [55] and Miller et al. [37]. In the former, it is shown that classifier performance is correlated with OOD detection accuracy, and in the latter that InD evaluations preserve ranking on OOD data. Taken together, these observations suggest a corollary that OOD detection rates are correlated with OOD accuracy and therefore generalization. We first validate this corollary experimentally by assessing the detection rates of several popular OOD detector implementations on several types and intensities of covariate shift, across multiple datasets, and measure the resulting relationship with generalization. The results reveal that there is frequently a strong relationship, which we operationalize into a simple accuracy estimator under covariate shift that does not require labels at inference time.

A key observation underlying this work is that OOD detectors are often criticized for responding strongly to covariate shifts and classifier-specific artifacts rather than purely to semantic novelty [20, 32, 35, 37, 43, 55, 61]. We take the opposite view: this behavior is useful. If detector verdict rates track the severity of covariate shift, then they implicitly encode information about the resulting degradation in model accuracy. We therefore treat aggregate OOD detection rates as a signal for estimating network accuracy, rather than a failure mode in semantic shift detection.

Specifically, we train a linear model which maps the detection rate of a given OOD detector over a given sequence of data to an accuracy estimate. We assess the resulting model in terms of five separate benchmarks, ten synthetic shifts, and six OOD detector implementations. We further contextualize our results against a baseline of simple hold-out evaluation, as well as the strongest competing methods for online accuracy estimation, namely Probabilistic Runtime Evaluation (PRE) [51] and Average Thresholded Confidence (ATC) [18]. Our results indicate that our approach provides superior accuracy estimates, ranging from 0.015 to 0.084 and outperforming competing methods by up to 75%. All code used as part of this work is available at our GitHub Repository¹.

We summarize our contributions as follows:

1. We reinterpret a known limitation of OOD detectors—their sensitivity to covariate shift and classifier-specific effects—as a useful signal. In particular, we show that aggregate detector verdict rates can serve as sequence-level statistics that correlate with shift-induced accuracy degradation.
2. We provide a systematic empirical characterization of this relationship across multiple datasets, architectures, detector types, and shift intensities, showing that OOD detection rates frequently exhibit a strong monotonic relationship with generalization gaps.
3. We operationalize this signal into a simple estimator of accuracy under covariate shift based on linear regression over detection rates, explicitly targeting sequence-level performance prediction, which does not require labels at inference time.
4. We demonstrate that this estimator consistently outperforms competing accuracy estimation baselines (ATC[18] and PRE[51]) across all evaluated datasets, achieving mean absolute errors between 0.02 and 0.10

2 Related Work

Distributional shift is a central threat to neural-network reliability [38]. We focus on covariate shift, where $P(x)$ changes while the task semantics are preserved, since this is the setting in which a deployed model may still be expected to perform but its measured InD accuracy may no longer be valid. This differs from label shift, which changes $P(y)$, and semantic shift, which changes the task or label space itself. These sorts of shift incur a significant risk of causing silent *generalization failure*, where the network fails to predict correctly in operational scenarios without yielding alerts, and is prevalent in a wide variety of domains [3, 5, 6, 9, 14, 19, 28, 41, 42, 58].

¹<https://anonymous.4open.science/r/GeneralizationPredictionCovariateShift-6A51>

Most prior work attempts to improve robustness through domain generalization, domain adaptation, uncertainty quantification, or large-scale multimodal pretraining [1, 17, 28, 46, 57]. Rather than attempting to increase model generalization, some have investigated methods by which the reliability of DNN outputs can be assured through other means. Notable examples of this include the implementation of abstention classes [49], such that DNN classifiers can defer predictions when there is insufficient basis for a reliable prediction. In LLM research, similar approaches have been proposed in order to reduce the incidence of hallucinations [27] or to avoid unsafe outputs [4, 25, 62]. Recent work has highlighted that OOD detectors may operationalize this approach in vision domains [52–54].

OOD detectors are intended to detect instances of data with near-zero density in the (learned) data distribution, i.e. $P(x, y) \approx 0$. There exists a wide variety of means by which this can be achieved, therein logit-based methods [22, 33, 34], gradient-based methods [23, 24, 30], density-based methods [8, 39, 40], representation-based methods [31, 48], and more [60]. Classification is then performed by setting a threshold according to known InD (and OOD) data [60].

Recent work questions whether standard OOD detectors primarily measure semantic novelty, covariate shift, classifier quality, or mixtures thereof [20, 32, 35, 37, 43, 55, 61]. This apparent coupling has in certain works been treated as useful rather than pathological: if detector verdict rates track covariate-shift severity, they can serve as stream-level statistics for prediction correctness assessments. This has for instance been leveraged to build models for the label-free prediction of DNN loss [54] and probabilistic models of network accuracy [51]. A separate line of work tackles label-free accuracy estimation directly from classifier confidence. ATC [18] thresholds a confidence score (max softmax or negative entropy) on the labeled InD validation set so that the resulting error rate matches the InD error rate, then estimates accuracy on a new batch as the fraction of inputs above the threshold. While the present work shares ATC’s goal, the operational signal is different: ATC aggregates per-sample confidence statistics, whereas we aggregate detection rates from dedicated OOD detectors that are explicitly designed to discriminate distributional support.

3 Methodology

In this section, we outline the methodology used for our analysis. We divide our experiments in two sections: in the first, we seek to assess the correlation between OOD detection rates and classifier generalization. In the second, we assess the degree to which a regression model built on this signal can provide accuracy estimates at competitive error rates with respect to existing baselines. Overall, we consider 5 different datasets and 6 different post-hoc OOD detector implementations. We also apply varying intensities of synthetic covariate shifts in addition to the organic covariate shifts already present in the datasets.

3.1 Datasets

We require datasets that contain organic instances of covariate shifts. To this end, we utilize several domain adaptation datasets, as well as a polyp segmentation benchmark consisting of four independently curated datasets. For the segmentation datasets, we define a correct prediction as a segmentation with Intersection over Union (IoU) greater than 0.5. An overview of these datasets is provided in Table 1.

3.2 Synthetic shifts

In addition to the natural covariate shifts provided in the aforementioned datasets, we also utilize synthetic shifts to assess how OOD detectors respond to different types and intensities of covariate shifts. We denote a dataset with a given intensity and shift type as a *fold*. For the synthetic shifts, we bootstrap new folds from the InD validation dataset and apply a given intensity and type of data augmentation. We vary augmentation intensity via parameter α spanning a range that induces measurable but non-destructive shifts. The augmentations are listed in Table 2.

3.3 OOD Detectors

We study a selection of OOD detectors based on the taxonomy of Yang et al. [60]. We narrow the scope to post-hoc methods. An overview of the studied OOD detectors is provided in Table 3. We

Table 1: Overview of Datasets.

<i>Dataset</i>	<i>No. Images</i>	<i>Resolution</i>	<i>No. Classes</i>	<i>Train/Test Domains</i>	<i>Description</i>
NICO [65]	230000	Varying	60	1/5	domain-adaptation classification dataset partitioned by a number of contexts.
Office-Home [56]	15500	Varying	65	1/3	A domain-adaptation classification dataset partitioned by modality
Office-31 [45]	4110	Varying	31	1/2	A domain-adaptation classification dataset partitioned by camera type
CCT [7]	243187	1024X747	15	65/65	a domain-adaptation benchmark consisting of images taken from different camera-traps.
Polyps [2, 10, 26, 47]	2720	Varying	1	1/3	Polyp segmentation datasets collected from different centres. We use Kvasir as the InD dataset.

Table 2: Overview of synthetic augmentations.

Shift	Functional Description	α Range
Additive noise	$x = x + \alpha U(-1, 1)$	$(0, 0.5]$
Multiplicative Noise	$x = x + \alpha U(-1, 1)x$	$(0, 0.5]$
Salt and Pepper Noise	$\begin{cases} x[U(0, 1) < \alpha/2] = 0 \\ x[U(0, 1) > 1 - \alpha/2] = 1 \end{cases}$	$(0, 0.125]$
FGSM [11]	$x = x + \alpha \text{sign}(\nabla_x(L(\theta, x, y)))$	$[0, 0.5]$
Hue	HueJitter(α)	$[0.05, 0.3]$
Brightness	BrightnessJitter(α)	$[0.05, 1]$
Contrast	ContrastJitter(α)	$(0, 1]$
Smear	GridDistortion(α)	$(0, 0.5]$
JPEG-compression	ImageCompression(α)	$(0, 75]$
Fog	RandomFog(α)	$(0, 0.5]$

135 calibrate the OOD detectors on the InD validation dataset, such that a positive verdict is returned if
 136 the feature value is beyond the 95th percentile.

137 3.4 Network training

138 For the purposes of generality, we perform assessments on several different architectures. We train
 139 a ResNet101[21], a vision-transformer [15] for all the classification datasets, and a DeepLabV3,
 140 UNet [44], and Segformer [59] for the polyp segmentation datasets. All models are pre-trained on
 141 ImageNet. The training configuration is summarized in Table 4.

142 3.5 Detector evaluation

143 We treat the OOD detectors not as a binary classifier, typical in other literature, but rather as an
 144 approximate measurement device. We therefore compute the *detection rate* (DR) observed over

Table 3: Overview of OOD detectors. Note: The expected-log-likelihood term from Nalisnick et al. [39] is constant w.r.t. x and is omitted as it does not affect the percentile-based threshold.

Detector	Formula	Description
Entropy [22]	$H(x) = -\sum_y p_\theta(y x) \log p_\theta(y x)$	Predictive entropy of the model output distribution.
GradNorm [23]	$\ \nabla_{W_L} \text{CE}(\text{softmax}(f_\theta(x)), \mathcal{U})\ _1$	ℓ_1 norm of the gradient of the cross-entropy between the softmax output and a uniform target $\mathcal{U}$, taken with respect to the final-layer weights W_L .
Energy [34]	$E(x) = -\log \sum_y e^{f_\theta(x)_y}$	Energy score, i.e. negative log-sum-exp of logits $f_\theta(x)$.
kNN [48]	$\min_{x_i \in \mathcal{D}_{\text{train}}} \ z(x) - z(x_i)\ _2$	Distance between test encoding $z(x)$ and its nearest neighbour $z_k(x)$ in the training set.
MSP [22]	$\max(p_\theta(y x))$	Maximum softmax score.
Typicality [39]	$-\log p_{\text{GLoW}}(x)$	Negative log-likelihood under a GLoW generative model trained on the InD data.

Table 4: Overview of Training Method.

Component	Choice
Framework	Pytorch Lightning [16].
Pretraining	Imagenet
Optimizer	Adam [29]
Scheduler	Cosine Annealing w/ Warm Restarts
Batch size	16
Data augmentation	Random flip, Random rotate
Initial learning Rate	1e-3
Model Selection	Early Stopping
Loss	Cross Entropy Loss / Jaccard loss

different folds of data, and how it relates to the generalization gap associated with a given fold. We expect that shifts of greater intensity as given by the α parameter induce a correspondingly greater generalization gap, and that they are more readily detected as OOD.

3.6 Accuracy estimation and baselines

We formalize the generalization gap of a fold (or sequence) S drawn from a distribution P_S as

$$\Delta(S) = \text{Acc}_{\text{InD}} - \mathbb{E}_{(x,y) \sim P_S} [\mathbf{1}\{f(x) = y\}], \quad (1)$$

where f is the trained classifier and Acc_{InD} is the accuracy measured on the held-out InD validation fold. $\Delta(S) \geq 0$ indicates degraded performance under shift.

Our estimator. We fit a *linear* regression model $\hat{\Delta}(S) = w \cdot \text{DR}(S) + b$ that maps the OOD detection rate observed on S to an estimated generalization gap. Linearity is imposed deliberately: since the overall detection rate and accuracy of a sequence are weighted sums of those of its constituent sub-sequences, a linear estimator preserves additivity over mixed batches. We use scikit-learn’s `LinearRegression` (ordinary least squares). We evaluate using a leave-one-shift-out protocol. Each fold corresponds to a distinct distribution (either a synthetic shift instance or an organic domain). For a held-out shift S , the regression model is trained on (DR, Δ) pairs from all other shifts. Importantly, organic shifts are never used as regression training targets when they are part of the evaluation set. This ensures that the estimator is evaluated on distributional conditions that were not observed during regression fitting. For synthetic shifts, each augmentation type and intensity combination is treated as a separate shift. When evaluating a given shift type (e.g., Gaussian noise), all instances of that

shift type are excluded from regression training to prevent interpolation across intensities of the same transformation.

The best (architecture, detector) configuration per dataset is selected by mean MAE on the calibration data, on the basis that training-set MAE offers a pre-deployment opportunity for OOD detector algorithm selection.

Baselines. We compare against three label-free accuracy estimators:

- **Naive:** predict $\hat{\Delta}(S) = 0$ for every fold, i.e. assume InD performance under shift.
- **ATC** [18]: average threshold confidence. Given a confidence score $s(x)$ for each input, ATC chooses a threshold τ on the labeled InD validation fold such that $\Pr[s(x) < \tau] = 1 - \text{Acc}_{\text{InD}}$, i.e. τ is the $(1 - \text{accuracy})$ quantile of the InD score distribution. The estimated accuracy on a new batch S is then $\Pr_{x \sim S}[s(x) \geq \tau]$. We instantiate ATC-MC using maximum softmax probability and ATC-NE using negative predictive entropy.
- **PRE** [51]: probabilistic runtime estimation. PRE estimates the rate of distributional shift from the OOD detector’s verdict trace and combines it with calibration accuracies on a held-out OOD partition to estimate accuracy on new sequences.

All three baselines are evaluated under the same per-fold protocol as our estimator and the best (architecture, detector/variant) configuration per dataset is selected for each method. For fairness, all methods are calibrated using the same labeled calibration shifts. In particular, PRE is calibrated on the same shifts used to train our regression model, and ATC thresholds are determined using the InD validation set only. No method has access to labels from the held-out test shift.

4 Results

In this section, we first evaluate the extent to which OOD detector verdicts are predictive of generalization. We observe that there is frequently a consistently positive but imperfect relationship between OOD detection rates and generalization gaps. We then exploit this relationship to build predictive models of generalization, which attain lower errors compared to competing approaches.

4.1 Shift Detectability and Generalization

Covariate shifts differ from semantic shifts in that they can manifest in different degrees of performance degradation. We therefore assess the frequency at which different covariate shifts, with differing induced generalization gaps, are detected. This was achieved by applying varying intensities of synthetic shifts to the InD validation fold and computing the resulting accuracy of the DNN. For brevity, Figure 1 presents the combination of OOD detector and network architecture with the strongest correlation for each dataset. A fairly strong positive relationship between the OOD detection rate for a given shift and the generalization gap it induces is evident across all five datasets.

To verify that this relationship is not an artifact of the cherry-picked best configuration, Figure 2 reports the distribution of $|\rho|$ (Spearman rank correlation between DR and Δ) across every (architecture, detector) configuration we evaluated. The median $|\rho|$ exceeds 0.5 on four out of five datasets and remains positive on every configuration we evaluated, which indicates that the DR–gap coupling is a property of the underlying mechanism rather than of any single combination of detector and architecture. Note that, because correlations are computed under leave-one-shift-out evaluation, these relationships reflect generalization across distinct distributions rather than interpolation within a single shift family. These results are consistent with the works of Miller et al. [37] and Vaze et al. [55], and the corollary that OOD detector rates correlate with OOD generalization.

These results should be interpreted empirically rather than mechanistically. Detector verdict rates appear to provide coarse sequence-level summaries of how strongly an input stream departs from the validation distribution. Because the same departures can also degrade classifier accuracy, detection rates and generalization gaps often vary together. However, this coupling is not guaranteed: some shifts may reduce accuracy while remaining weakly detectable, whereas others may be strongly detected without causing proportional accuracy loss. The estimator introduced below therefore relies on observed monotonicity within a deployment setting, which may or may not hold in practice. This is discussed further in Section 5.

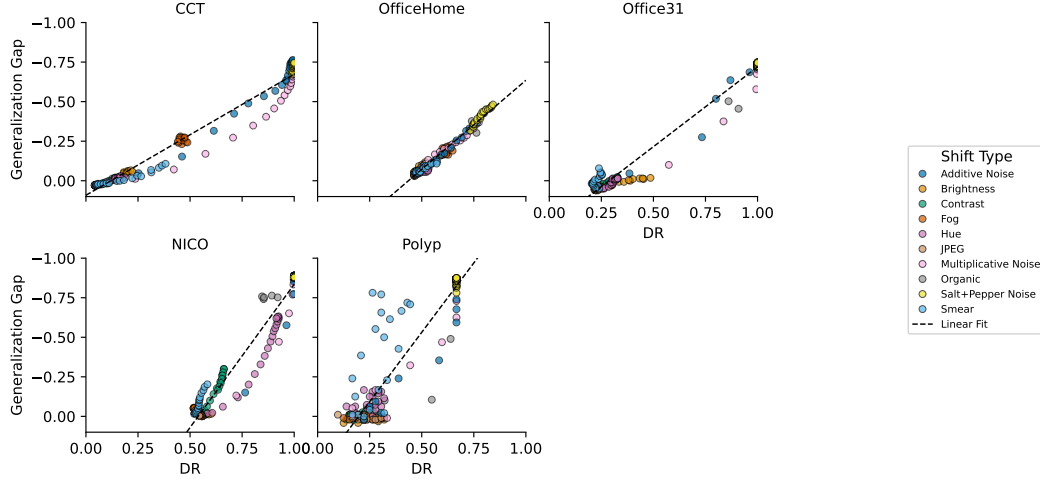


Figure 1: Strongest relationships between OOD detection rate and generalization gap across covariate shifts. Marker color encodes the shift type.

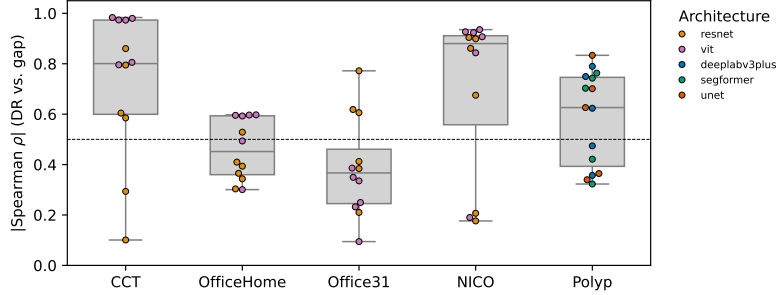


Figure 2: Distribution of the absolute Spearman correlation between OOD detection rate and the induced generalization gap, computed for every (architecture, detector) combination we evaluated. Each point is one configuration; the dashed line marks $|\rho| = 0.5$.

4.2 Accuracy Prediction

We exploit the observed relationship between OOD detection rate and generalization gap to construct a predictive model of accuracy under covariate shift. Using the protocol described in Section 3.6, we evaluate a linear regression estimator against three label-free baselines: a naive InD predictor, ATC [18], and PRE [51]. All methods are assessed under the same leave-one-shift-out setup, with the best configuration selected per dataset. Table 5 reports the resulting Mean Average Error (MAE).

	CCT	OfficeHome	Office31	NICO	Polyp
ATC-MC [18]	0.12	0.06	0.05	0.05	–
ATC-NE [18]	0.12	0.11	0.12	0.16	0.11
PRE [51]	0.07	0.08	0.14	0.17	0.24
Ours	0.04	0.02	0.03	0.05	0.10

Table 5: Per-dataset MAE for label-free accuracy estimation under per-fold leave-one-out evaluation. *Ours* is our linear regression of generalization gap on detection rate; ATC-MC denotes ATC with maximum-softmax confidence; ATC-NE denotes ATC with negative entropy. ATC-MC is undefined for Polyp segmentation outputs. Best per dataset in bold.

Our estimator attains the lowest MAE on every dataset, ranging from 0.02 (OfficeHome) to 0.10 (Polyp), and improves over the strongest baseline by 27–75%. This indicates that our approach

provides significantly more reliable estimates of accuracy under a wider range of conditions than the competing approaches. This indicates that the proposed estimator provides more reliable accuracy estimates across a range of shifts.

Estimator Comparisons Table 5 aggregates over all held-out folds. To verify that no single class of shift drives the result, Figure 3 disaggregates per-fold MAE along the synthetic-shift intensity axis (lines) and along organic shifts (markers, right of each panel). Our approach dominates ATC across the full intensity sweep on every dataset, and is competitive with or better than PRE across the organic shifts on four out of five datasets. The Polyp datasets represent the only setting where the three estimators converge. It is inconclusive whether this is a property of segmentation-specific calibration challenges that affect all label-free estimators similarly, or an artifact of the datasets themselves.

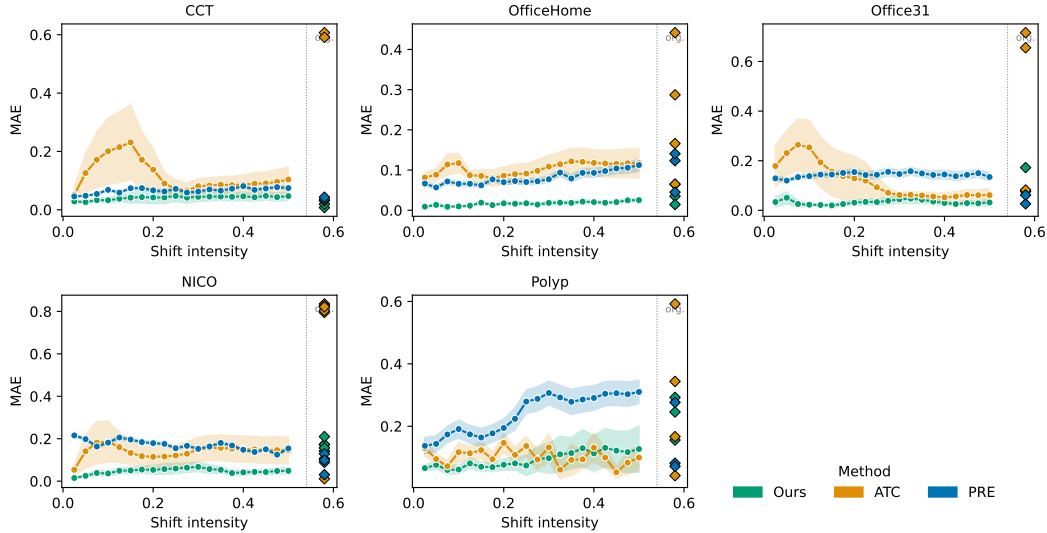


Figure 3: Per-fold MAE across synthetic-shift intensity (lines, with bands across shift types) and organic shifts (diamonds, right of dotted separator) for our estimator, ATC, and PRE. Our approach dominates ATC across the entire synthetic-intensity sweep on every dataset and is competitive with PRE on organic shifts.

Estimator Analysis Figure 4 plots per-prediction MAE as a function of the true generalization gap. The estimation errors are weakly correlated with generalization, but vary by orders of magnitude, particularly for shifts that induce only minor gaps where the estimator’s residual can exceed the gap itself (though remaining small in absolute terms). The Q–Q insets indicate that residuals are approximately symmetric but exhibit heavier-than-Gaussian tails on most datasets. A calibration analysis was also performed that revealed that our approach is significantly more well-calibrated than the competing approaches. This calibration analysis can be found in Section A. Finally, as our approach operates on a sequence-level, we performed a sensitivity analysis with respect to sequence length, which revealed an inverse relationship between MAE and sequence length and that low errors can be achieved with a sequence lengths as low as $n = 10$. This analysis can be found in Section B.

5 Discussion

Overall, our results support a consistent empirical coupling between OOD detection rates and generalization under covariate shift, which can be exploited for unlabeled deployment-time accuracy estimation.

However, these results should not be interpreted as evidence that OOD detectors universally solve silent generalization failure. Rather, they support a narrower claim: when detector scores vary approximately monotonically with shift-induced degradation, aggregate detection rates provide a useful signal for stream-level reliability estimation. This interpretation is consistent with recent critiques showing that supervised OOD detectors often respond strongly to covariate rather than semantic

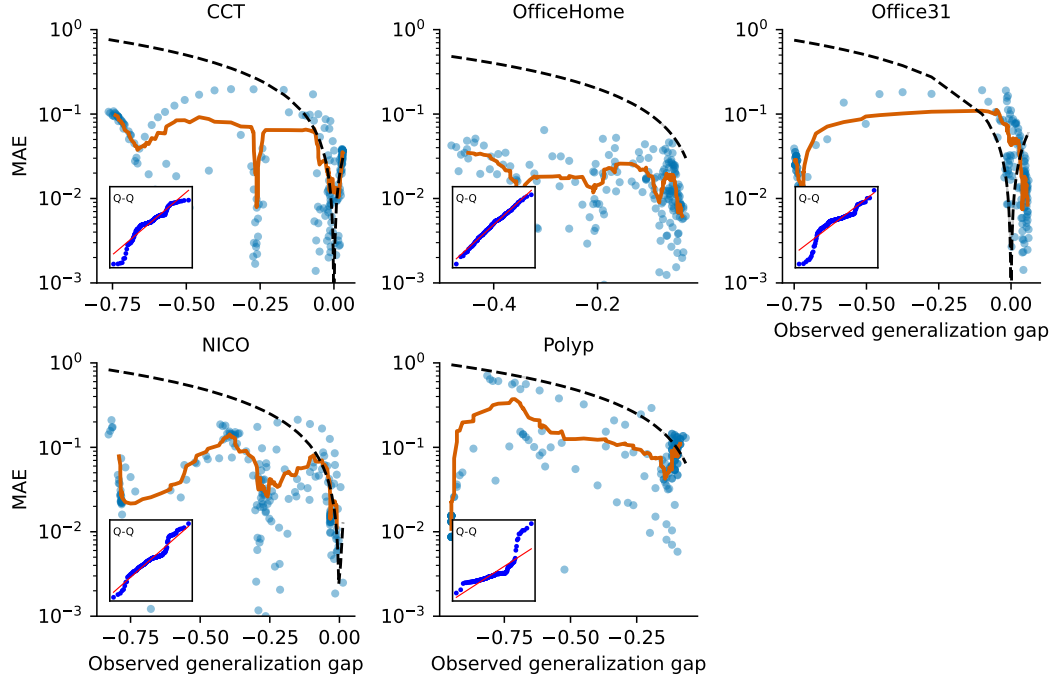


Figure 4: Per-prediction MAE as a function of the true generalization gap. The black dashed line is the naive baseline; the orange line is a sliding-window average of our estimator’s MAE. Insets show normal Q-Q plots of the residuals.

shifts [32, 61]. Importantly, the strength of this relationship may vary across detector architectures and shift types, and may break under certain conditions (see Figure 1). A broader evaluation across detectors, datasets, and shift types is therefore needed to characterize when the DR-gap relationship holds. In addition, while the estimator achieves low average error, its calibration varies across datasets, with some settings exhibiting mild bias or variance differences (see Appendix A).

We have in this work operated at the sequence level rather than per-sample, yielding coarse-grained estimates whose reliability may depend on sequence length. While detection rates track accuracy at the aggregate level, they do not necessarily identify individual errors. This is consistent with prior work showing that OOD detectors typically only weakly separate correct and incorrect predictions at the sample level, if at all [50].

Finally, our use of relatively simple synthetic shifts may limit coverage of more complex real-world distribution changes. Evaluating the approach under richer, model-driven or naturally occurring shifts is an important direction for future work. Moreover, while we have intentionally opted for a simple augmentation pipeline in this work to facilitate the construction of synthetic datasets with a sufficient effect size, this may impact the applicability of our results to engineering settings. The aforementioned improvements would provide the secondary benefit of permitting more realistic augmentation pipelines, including RandAugment[13], MixUp[64], and CutMix[63].

6 Conclusion

In this paper, we have shown that aggregate OOD detection rates provide a simple and effective signal for predicting accuracy under covariate shift without labels. Across multiple datasets, detectors, and shift types, a linear estimator based on detection rates consistently outperformed competing approaches. These results suggest that OOD detectors, despite known limitations at the sample level, can serve as useful sequence-level monitoring signals in deployed systems. An important direction for future work is to identify conditions under which the DR-gap relationship is reliable and to design detectors that explicitly optimize for this coupling.

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A Calibration Analysis of Accuracy Estimators

In addition to mean absolute error, we evaluate calibration by comparing predicted and observed generalization gaps. Figure 5 presents predicted versus observed gaps for each method and dataset, along with summary statistics (bias, MAE, and slope). The dashed line indicates perfect calibration, while the solid line shows a linear fit.

Across datasets, our estimator is consistently well-calibrated, with near-zero bias and slopes close to one. Specifically, bias remains within ± 0.006 on CCT, OfficeHome, Office31, and NICO, and slightly negative on Polyp (-0.005), indicating minimal systematic over- or underestimation. Slopes range from 0.90 to 0.98 on most datasets, reflecting good alignment between predicted and observed gaps, with a modest underestimation tendency on Polyp (slope 0.83). MAE is also consistently low (e.g., 0.017 on OfficeHome and 0.034 on Office31), confirming strong calibration and accuracy.

In contrast, ATC-MC exhibits substantially higher bias (up to $+0.089$ on CCT and $+0.059$ on OfficeHome) and pronounced slope compression (e.g., 0.41 on CCT and ≈ 0.5 on OfficeHome and Office31), indicating systematic underestimation of large gaps and overestimation of small ones. While its MAE is competitive on some datasets (e.g., 0.047 on NICO), its calibration is less stable across shifts.

ATC-NE shows the largest variability and bias overall. Bias exceeds $+0.09$ on multiple datasets and reaches $+0.148$ on NICO, with MAE as high as 0.158. Slopes are inconsistent, ranging from underconfident (0.71 on Office31) to overconfident (1.32 on OfficeHome), indicating unreliable scaling of predictions.

Overall, the results indicate that our method achieves both low error and strong calibration across datasets, with only mild slope compression in more challenging settings (e.g., Polyp). Competing methods exhibit either systematic bias, slope distortion, or instability across datasets, reinforcing the robustness advantages of our approach.

B Sequence Length Sensitivity Analysis

We evaluate how predictor MAE varies with respect to sequence length.

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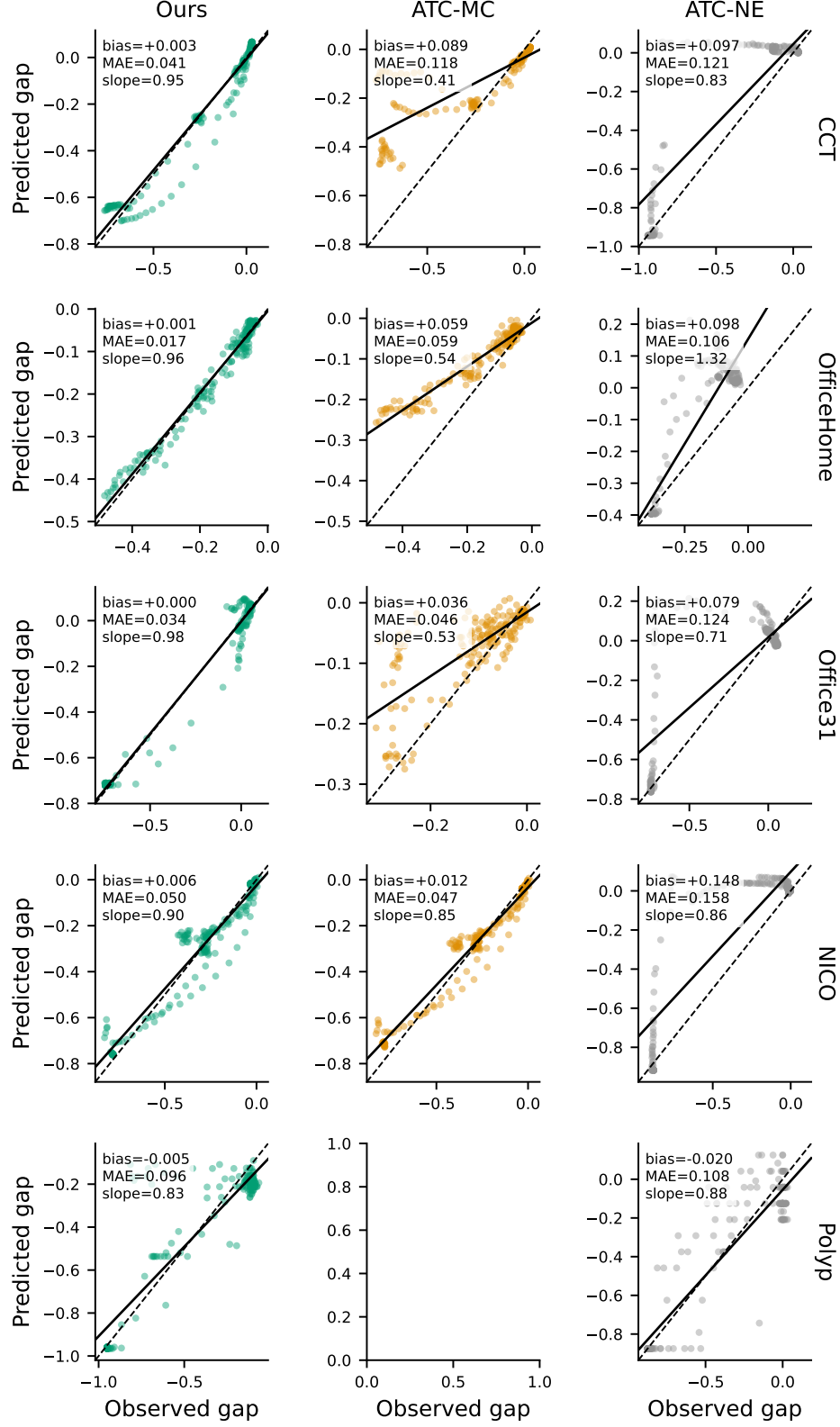


Figure 5: Predicted versus observed generalization gaps for each method (rows) and dataset (columns). The dashed line indicates perfect calibration; the solid line is a linear fit. Annotations report bias, MAE, and slope.

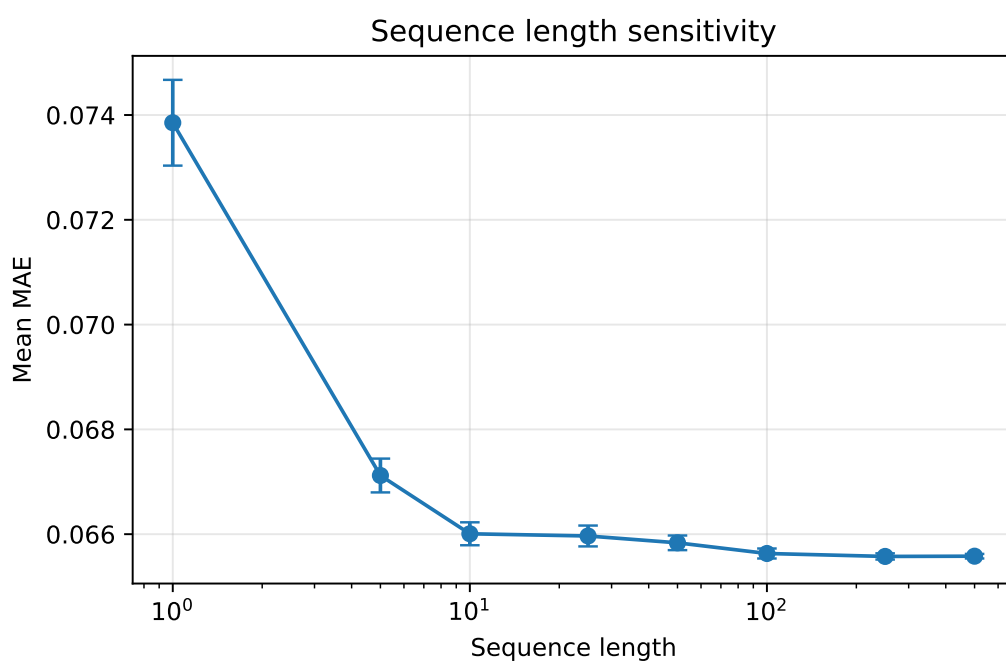


Figure 6: Sensitivity analysis of MAEs with respect to sequence length. MAE is approximately inversely proportional to sequence length, with low errors being possible with as little as sequences of length $n = 10$.